

COURSE TITLE : WOOD PRODUCTS III
COURSE CODE : 6123
COURSE CATEGORY : A
PERIODS PER WEEK : 5
SEMESTER : 6
PERIODS PER SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1.	Fibreboard	5
	Chipping	5
	Pulping	7
	Test	1
2.	Mat formation	5
	Insulation Board	5
	Hardboard	5
	Test	1
3.	Oriented strand board	5
	Cement bonded PB	5
	Dimensional stabilisation	8
	Test	1
4.	Sandwich composition	10
	Bulking Treatment	6
	Testing and Properties of Wood laminates	5
	Test	1
	Total	75

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand fibreboard
	2	Comprehend process for fibreboard manufacture
2	1	Understand the mat formation in fibreboard
	2	Comprehend insulation board
	3	Understand the production of hard board
3	1	Understand orient strand board
	2	Know cement bonded particleboard

	3	Understand dimensional stabilisation
4	1	Comprehend sandwich composition
	2	Understand bulking treatment

Specific Outcome

MODULE-I

1.1.0 Understand Fibreboard

- 1.1.1 Define fiber board insulation board MDF and hard board
- 1.1.2 State the history and development of fiber board
- 1.1.3 State the classifications of fiber board
- 1.1.4 State the modified wood raw materials
- 1.1.5 Describe the non wood fibrous raw materials
- 1.1.6 Explain properties and uses of different types of fiber board
- 1.1.7 List the properties of semi rigid insulation board, rigid insulation board, different type of hard boards
- 1.1.8 Describe the schematic outline of fiber board manufacturer using wet, semi dry and dry process.

1.2.0 Comprehend Process For Fibreboard Manufacture

- 1.2.1 State the need of chipping
- 1.2.2 Describe different types of chippers used in FB manufacture
- 1.2.3 Describe magnetic separation and screening of fibers
- 1.2.4 State storage of chips and the importance of the M.C of chips during storage.
- 1.2.5 State the need of wetting of chips.
- 1.2.6 Describe the moisture content of chip before pulping
- 1.2.7 List the four basic pulping processes.
- 1.2.8 Describe ground wood pulping by pocket grinders.
- 1.2.9 Describe ground wood pulping by magazine type grinders.
- 1.2.10 Describe chemical pulping by rotary drum digester
- 1.2.11 Explain thermo mechanical process

MODULE-II

2.1.0 Understand the Mat Formation in Fibreboard

- 2.1.1 State the need of mat formation
- 2.1.2 Describe the importance of mat formation.
- 2.1.3 List methods of mat formation.
- 2.1.4 Explain the mat formation by deckle box.
- 2.1.5 Describe the mat formation by cylinder method,,Fourdrinier method

2.2.0 Comprehend Insulation Board

- 2.2.1 Define insulation board
- 2.2.2 Draw and describe the block diagram of manufacture
- 2.2.3 Describe sizing of in insulation stock.
- 2.2.4 Describe the mat formation
- 2.2.5 Describe drying
- 2.2.6 Describe trimming and finishing.

2.3.0 Understand the Production of Hard Board

- 2.3.1 Define Fibreboard
- 2.3.2 Draw the block diagram of hardboard manufactures.
- 2.3.3 Describe the sizing.
- 2.3.4 Describe the mat formation and Pressing.
- 2.3.5 Describe oil tempering, heat treatment, Humidification.
- 2.3.6 Describe testing methods of fiber board.
- 2.3.7 State the standard test results of M D F, Hardboard.

MODULE-III

3.1.0 Understand Orient Strandboard

- 3.1.1 Define Oriented Strand board
- 3.1.2 Describe Manufacturing Methods
- 3.1.3 Define wood wool board
- 3.1.4 Describe the Method of manufacture of wood wool board
- 3.1.5 Describe its Properties and uses

3.2.0 Know Cement Bonded Particleboard

- 3.2.1 Define Cement Bonded Particleboard
- 3.2.2 Describe the Method of Manufacture
- 3.2.3 Describe the Properties and uses of CBPB
- 3.2.4 Explain Moulded particle Board
- 3.2.5 Describe Werzalit Process

3.3.0 Understand Dimensional Stabilisation

- 3.3.1 State Densified Wood
- 3.3.2 Explain the method of Preparation of raw materials
- 3.3.3 Explain the process for densification of Wood by Impregnation and Compression
- 3.3.4 State Metalized Wood
- 3.3.5 Describe the Manufacture of Compreg
- 3.3.6 Define Dimensional stabilisation of wood
- 3.3.7 Describe the various methods of dimensional stabilisation such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking.

MODULE IV

4.1.0 Comprehend Sandwich Composition

- 4.1.1 Define sandwich Composition
- 4.1.2 Explain the raw materials
- 4.1.3 Describe Method of Manufacture
- 4.1.4 Describe Edge banding of Particle Board
- 4.1.5 Describe Laminated PB

4.2.0 Understand Bulking Treatment

- 4.2.1 Define Bulking treatment
- 4.2.2 List bulking treatments
- 4.2.3 Explain Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment
- 4.2.4 Explain Properties and uses of impreg
- 4.2.5 Explain acetylated wood
- 4.2.6 Describe Testing and Properties

CONTENT DETAILS

MODULE I

Fiber board-introduction, ISO and FAO definitions, classification, history and development. Properties and uses of different types of fiber boards- rigid insulation board, semi rigid insulation board, standard hard board, medium density hard board and special density hard boards. General outline of fiber board manufacture using wet, semidry and dry process. Modified wood raw materials and non-wood fibrous raw materials. Debarking of wood- (-brief descriptions)

MODULE II

Chipping of wood, need of chipping, different types of chippers (list and brief ideas only). Screening, storing. Chip moisture content, wetting of chips. Pulping process-ground wood pulping- chemical pulping, explosion or masonite process, thermal - mechanical or defibrator process.chemical-mechanical pulping process mat formation- Deckle box method, cylinder method, fourdrinier method. Block diagram of insulation board manufacture, sizing, mat formation drying, trimming and finishing. Block diagram of hard board manufacture, sizing, mat formation, pressing, trimming, finishing. Properties, testing and uses of hard board and MDF. Standard test results of MDF and hardboard. oil tempering, heat treatment, Humidification. testing methods of fiber board..

MODULE-III

Oriented Strand board Manufacturing Methods, wood wool board, the Method of manufacture of wood wool board, its Properties and uses, Cement Bonded Particleboard, the Method of Manufacture ,the Properties and uses of CBPB, Moulded, particle Board, Werzalit Process dimensional stabilisation, Densified Wood, the method of Preparation of raw materials, the process for densification of Wood by Impregnation and Compression, Metalized Wood, the Manufacture of Compreg, Dimensional stabilisation of, wood, the various methods of dimensional stabilisation such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation &cross linking.

MODULE IV

sand witch Composition, the raw materials, Method of Manufacture, Edge banding of Particle Board, Laminated PB, Bulking treatment, bulking treatments, Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment, Properties and uses of impreg, acetylated wood, Testing and Properties

REFERENCE

1. Franz F P Kollmann, PRINCIPLES OF WOOD SCIENCE AND TECHNOLOGY-VOLUME-II ,Springer Verlag
- 2 MODERN PARTICAL BOARD AND DRYING PROCESS OF FIBERBOARD